



Data Communications and Networking

Fourth Edition

Chapter 9

Using Telephone and Cable Networks for Data Transmission



9.1 TELEPHONE NETWORK

Telephone networks use circuit switching. The telephone network had its beginnings in the late 1800s. The entire network, which is referred to as the **plain old telephone system (POTS)**, was originally an analog system using analog signals to transmit voice.

Topics discussed in this section:

Major Components

LATAs

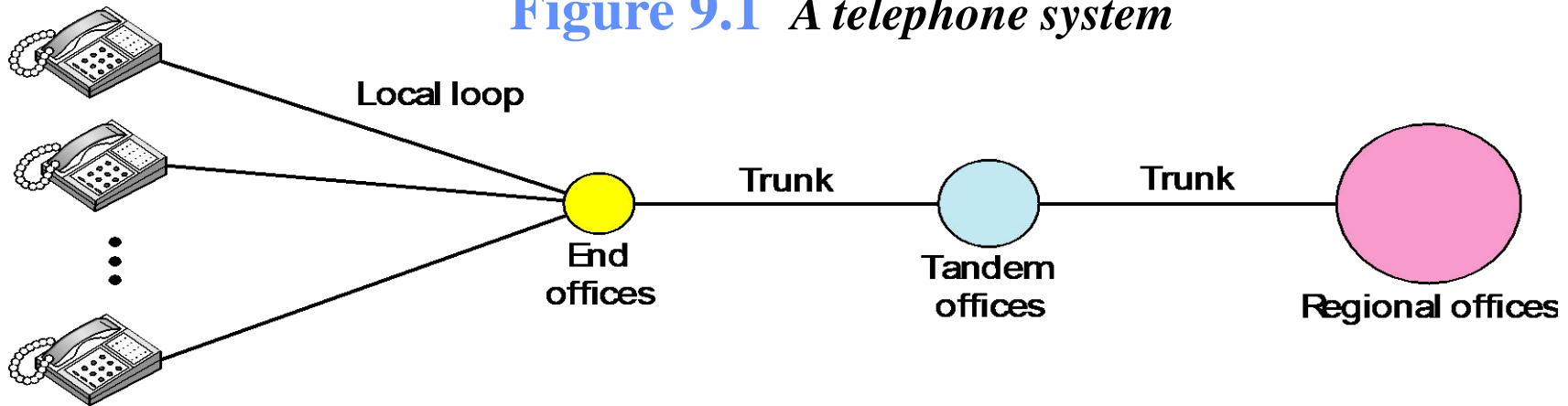
Signaling

Services Provided by Telephone Networks



Telephone Networks : Major Components

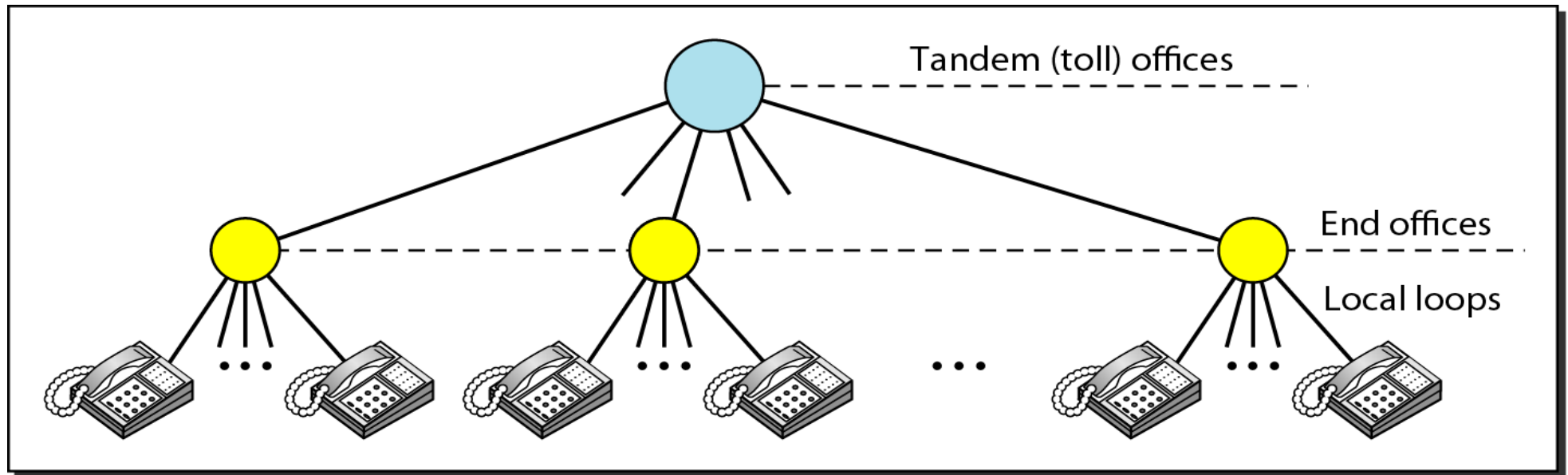
Figure 9.1 *A telephone system*



- local loop connects subscriber to nearest end office, 1st 3 digits of phone number define the office, next four define the local loop number
- trunks are the transmission media handling the communication between offices
- switching office has switches that connects several local loops or trunks

Telephone Networks

Figure 9.2 *Switching offices in a LATA*

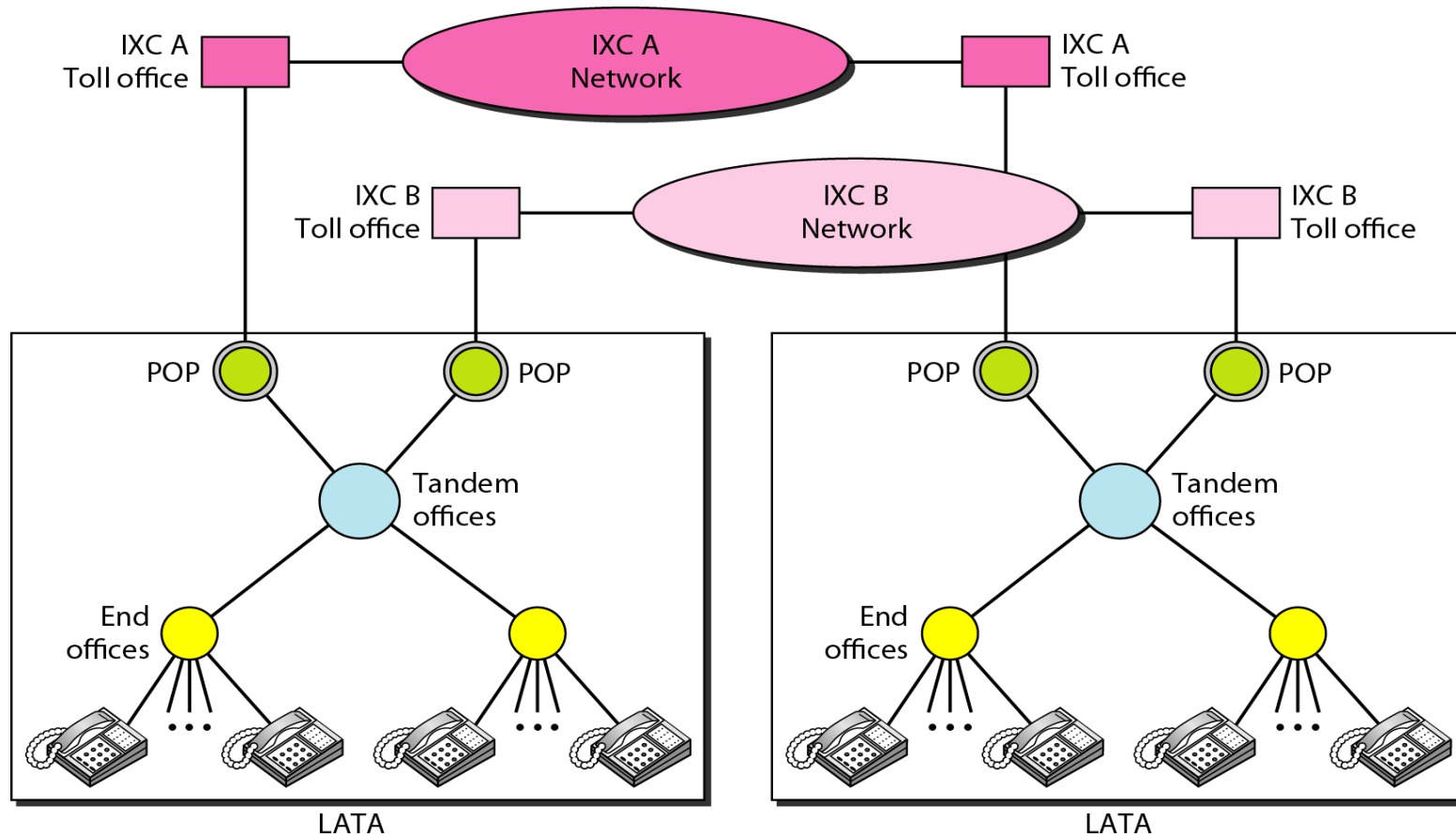


LATA (*Local –access Transport area*)



Telephone Networks

Figure 9.3 *Point of presences (POPs)*



Signaling

❑ The switches in the telephone company used the digital signals (telephone number) to create a connection between the caller and the called parties.

❖ In-band signaling

- The 4khz voice channel is used to provide signaling.
- The same circuit can be used for both signaling and voice communication.

❖ Out-band signaling

- A portion of the 4khz voice channel bandwidth is used for signaling.
- The voice bandwidth and the signaling bandwidth are separated.



Signaling

Note

**The tasks of data transfer and signaling are separated in modern telephone networks:
data transfer is done by one network,
signaling by another.**



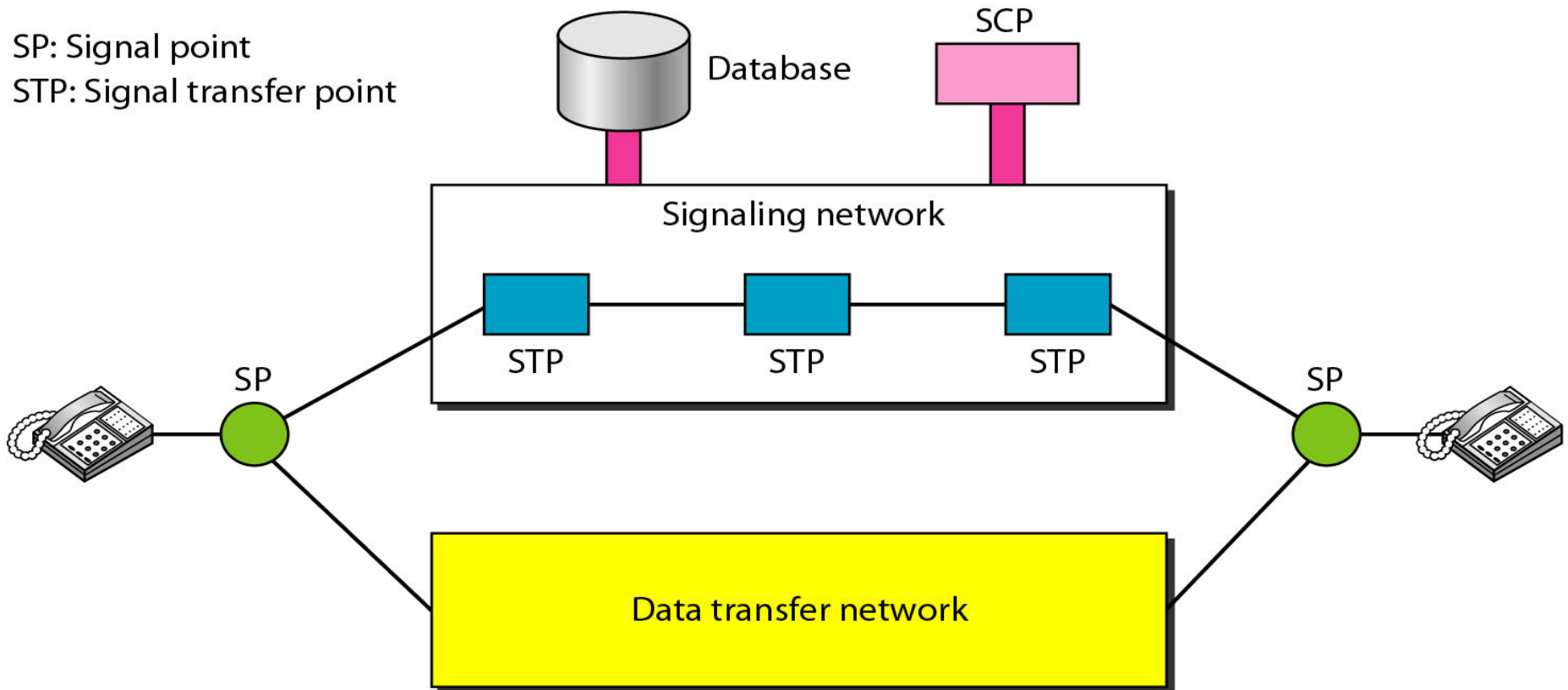
Signaling Network

- ❑ The signaling network is a packet-switched network involving the layers in the OSI model or Internet model.
- ❑ For example, the information needed to convey a telephone address can easily be encapsulated in a packet with all the error control and addressing information.
- ❑ The signaling network is consisted of the SP, STP, SCP, and Data base.



Signaling Network

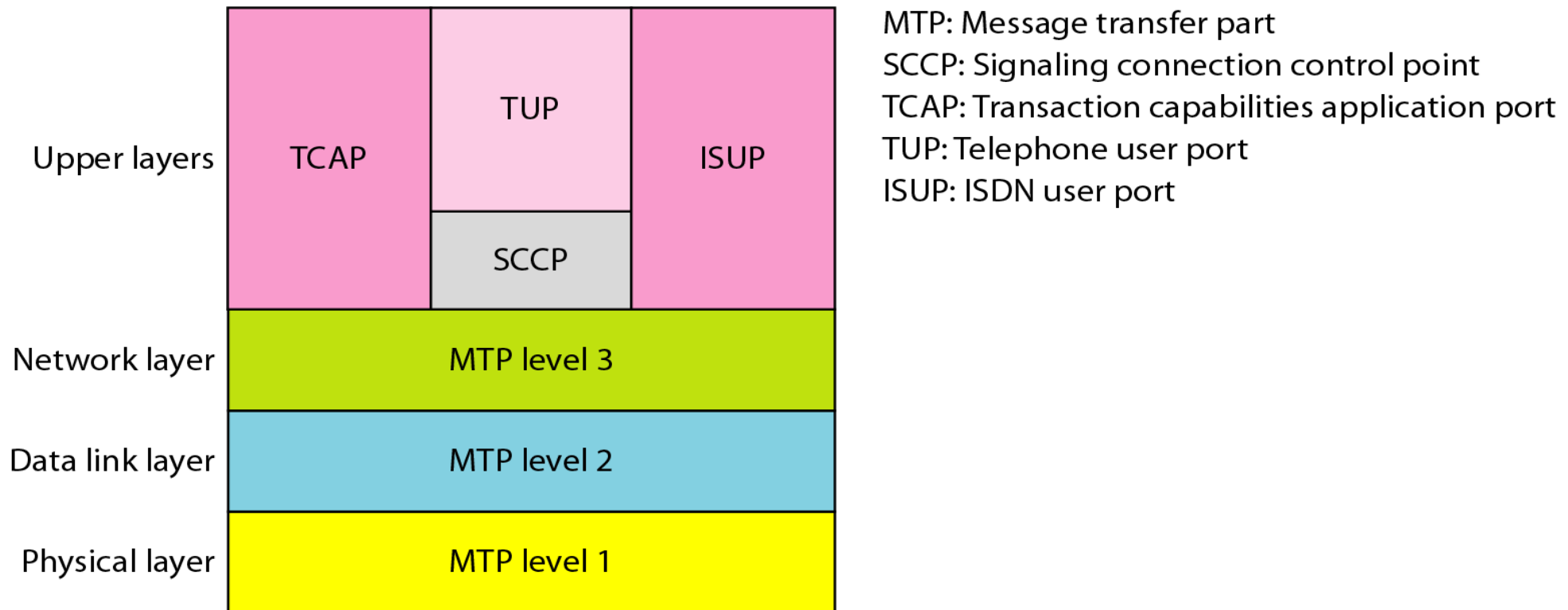
Figure 9.4 *Data transfer and signaling networks*



Signaling system seven (SS7)

- ❑ The protocol that is used in the signaling network is called Signaling System Seven (SS7)

Figure 9.5 *Layers in SS7*



9.2 DIAL-UP MODEMS

Traditional telephone lines can carry frequencies between 300 and 3300 Hz, giving them a bandwidth of 3000 Hz. All this range is used for transmitting voice, where a great deal of interference and distortion can be accepted without loss of intelligibility.

Topics discussed in this section:

Modem Standards



Dial-up modem

- ❑ Traditional, telephone lines can carry frequencies between 300 and 3300Hz, giving them a bandwidth of 3000 Hz.
- ❑ The effective bandwidth of a telephone line being used for data communication is 2400 Hz, covering the range from 600 and 3000 Hz.

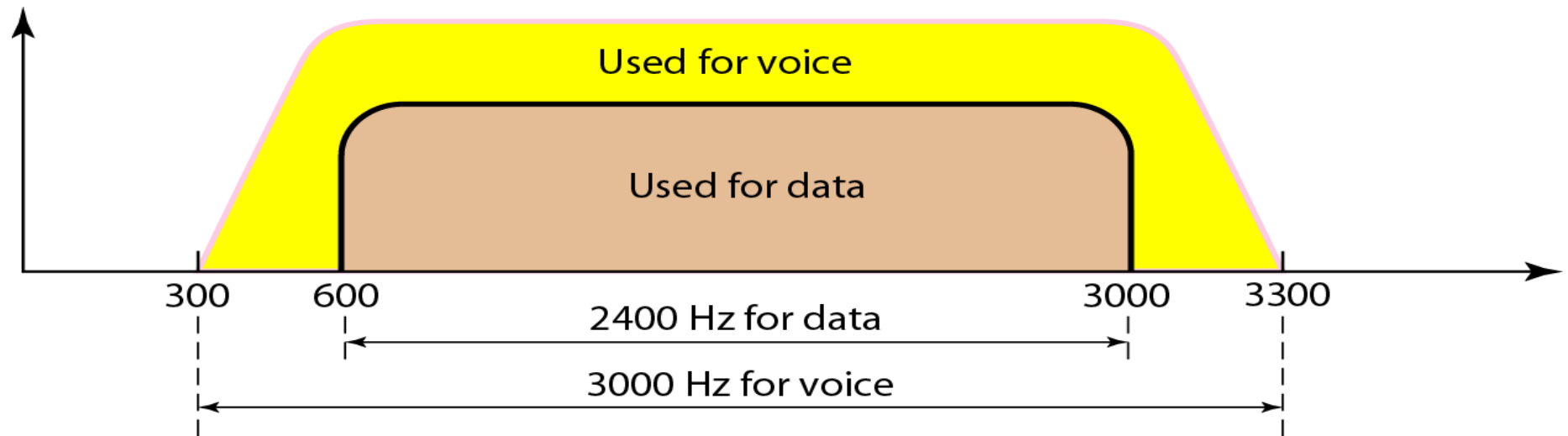


Figure 9.6 *Telephone line bandwidth*

Dial-up modem

Note

Modem

stands for modulator/demodulator.



Modem

- ❑ The computer sends a digital signal to the modulator portion of the modem; the data sent as an analog signal on the telephone lines.
- ❑ The modem on the right receives the analog signal, demodulates it through its demodulator, and delivers data to the computer on the right.

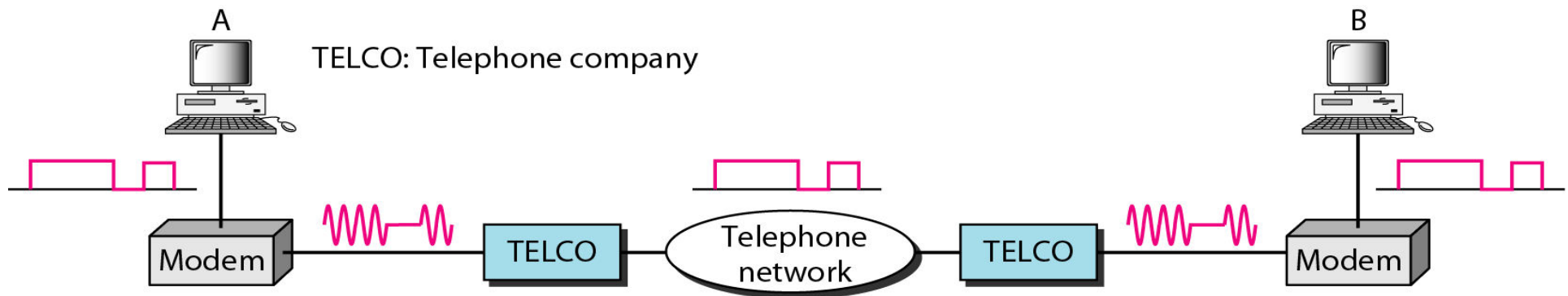


Figure 9.7 *Modulation/demodulation*

Modem standards

□ ITU V Series

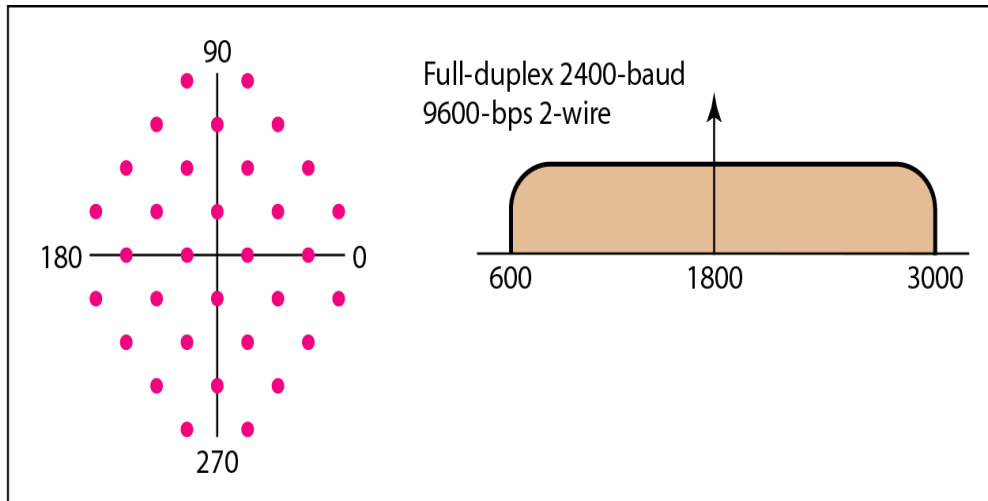
□ 32-QAM (V.32) with a baud rate of 2400

□ 4 data bits x 2400
= 9600 bps

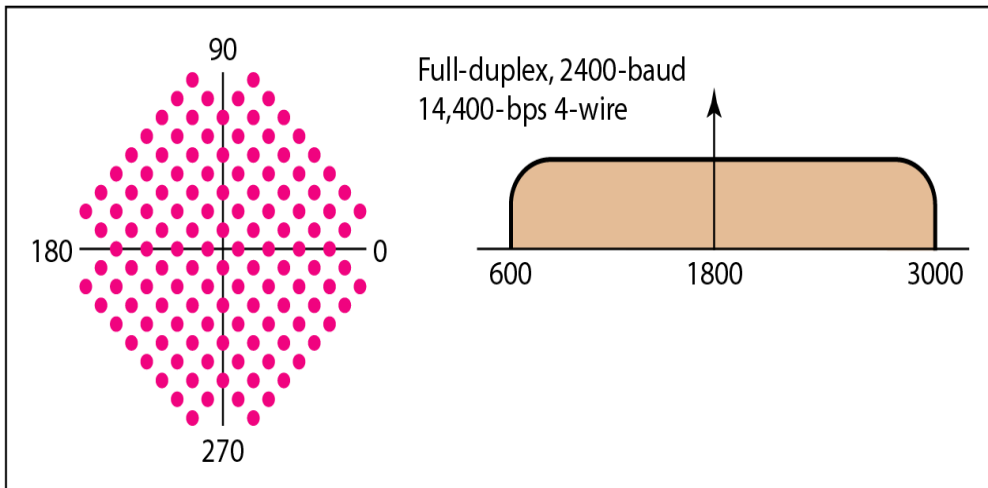
□ 128-QAM (V.32bis)

□ 6 data bits x 2400 baud
= 14,400 bps

□ Automatic fall-back, fall-forward feature enabling modem to adjust speed depending on line or signal quality



a. Constellation and bandwidth for V.32

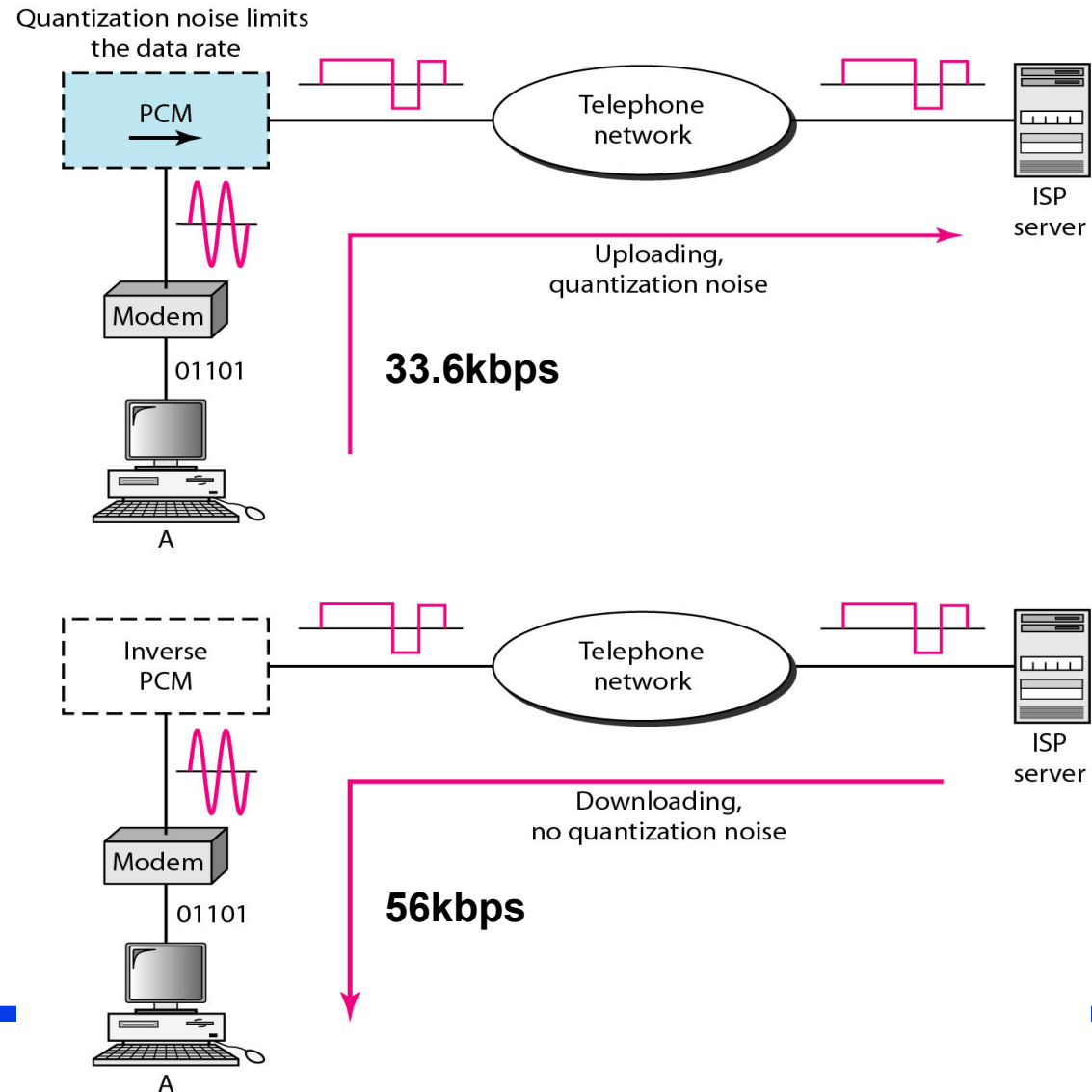


b. Constellation and bandwidth for V.32bis



Modem standards

Figure 9.9 *Uploading and downloading in 56K modems*



9.3 DIGITAL SUBSCRIBER LINE

After traditional modems reached their peak data rate, telephone companies developed another technology, DSL, to provide higher-speed access to the Internet. **Digital subscriber line (DSL)** technology is one of the most promising for supporting high-speed digital communication over the existing local loops.

Topics discussed in this section:

ADSL

ADSL Lite

HDSL

SDSL

VDSL



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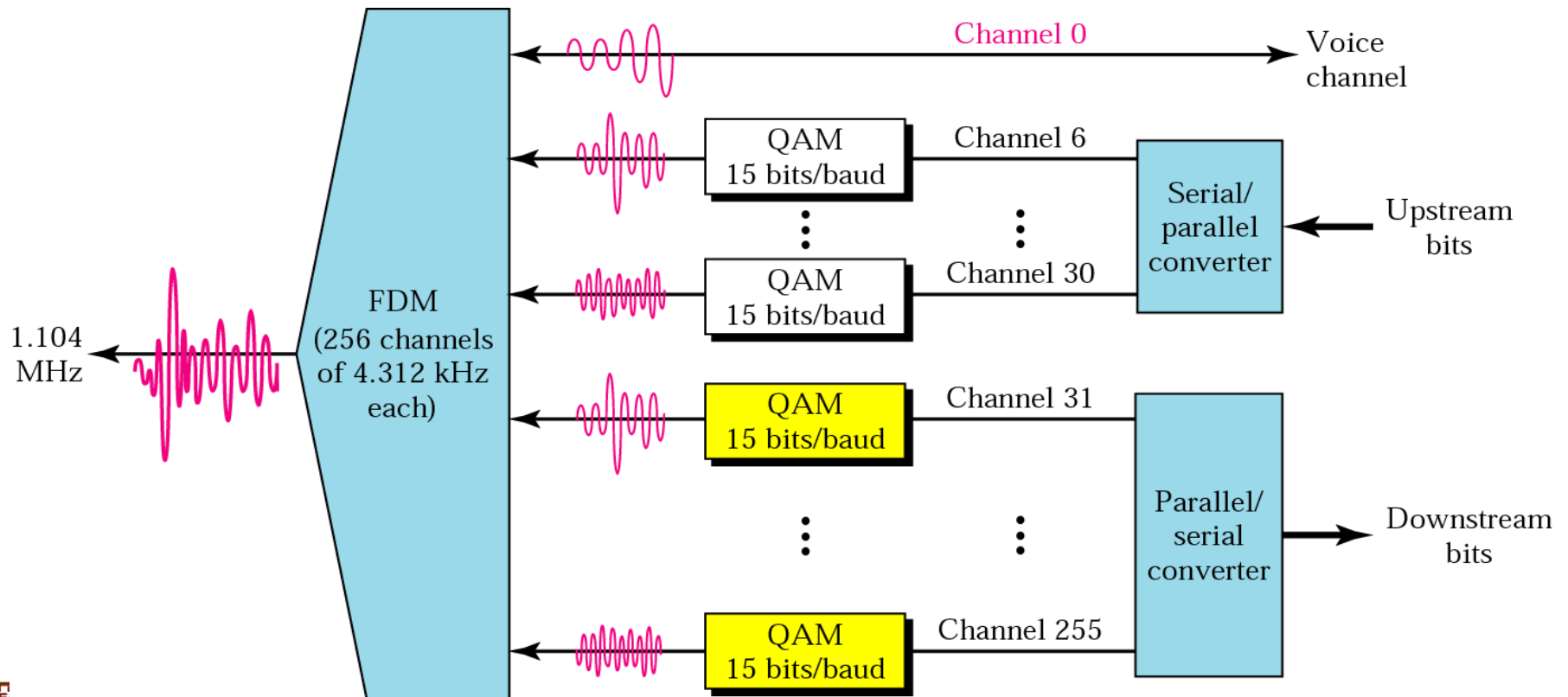
Digital Subscriber Line (DSL)

- ❑ uses a newer technology that used the existing telecommunications networks such as the local loop telephone line.
- ❑ is an asymmetric communication technology designed for residential users; it is not suitable for business.
- ❑ xDSL: where x can be replaced by A, V, H, or S
- ❑ The existing local loops can handle bandwidths up to 1.1 MHz
 - ❖ by removing the filter at the end of line of telephone company
 - ❖ but, limitation because of distance between the residence and the switching office, size of cable
- ❑ ADSL is an adaptive technology. The system uses a data rate based on the condition of the local loop line

DSL

DMT

- ❖ Modulation technique that has become standard for ADSL is called the discrete multitone technique (DMT) which combines QAM and FDM.



DSL

- ❑ **voice : channel 0 is reserved for voice**
- ❑ **Idle : channel 1 to 5 are not used; gap between voice and data communication**
- ❑ **Upstream data and control : channels 6 to 30 (25channels); one channel for control**
- ❑ **Downstream data and control : channels 31 to 255(225 channels); 13.4 Mbps; one channel for control**

DSL

Figure 9.11 *Bandwidth division in ADSL*

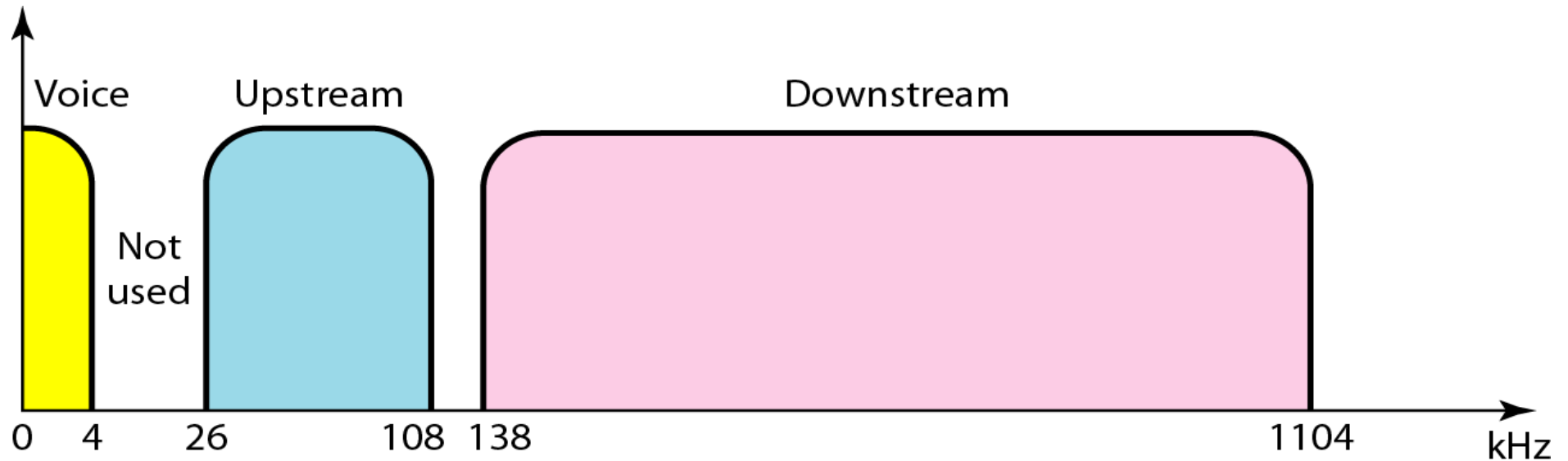
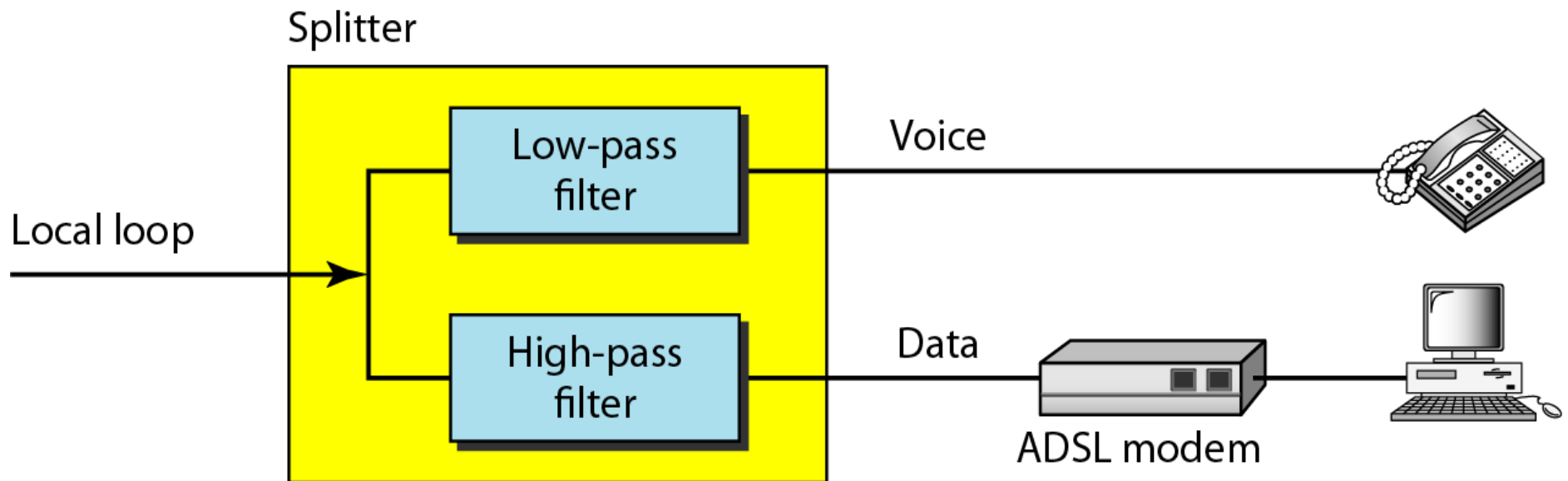
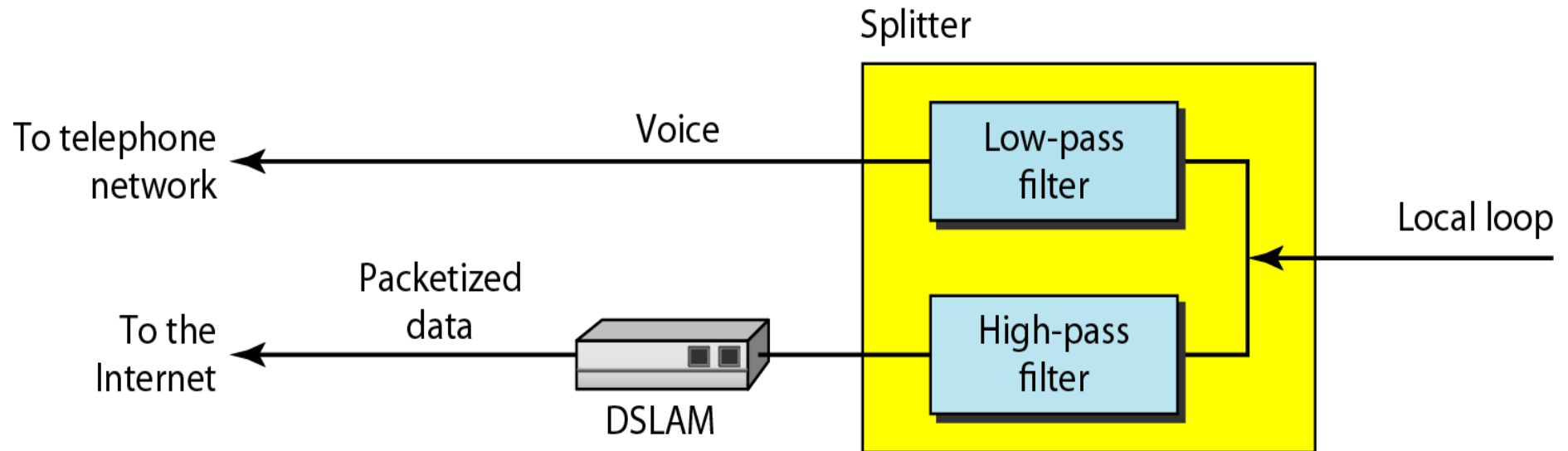


Figure 9.12 *ADSL modem – customer site*



DSL

Figure 9.13 *DSLAM – Telephone company site*



DSLAM (Digital Subscriber Line Access Multiplexer)

DSL

Table 9.2 *Summary of DSL technologies*

<i>Technology</i>	<i>Downstream Rate</i>	<i>Upstream Rate</i>	<i>Distance (ft)</i>	<i>Twisted Pairs</i>	<i>Line Code</i>
ADSL	1.5–6.1 Mbps	16–640 kbps	12,000	1	DMT
ADSL Lite	1.5 Mbps	500 kbps	18,000	1	DMT
HDSL	1.5–2.0 Mbps	1.5–2.0 Mbps	12,000	2	2B1Q
SDSL	768 kbps	768 kbps	12,000	1	2B1Q
VDSL	25–55 Mbps	3.2 Mbps	3000–10,000	1	DMT



9.4 CABLE TV NETWORKS

The **cable TV network** started as a video service provider, but it has moved to the business of Internet access. In this section, we discuss cable TV networks per se; in Section 9.5 we discuss how this network can be used to provide high-speed access to the Internet.

Topics discussed in this section:

Traditional Cable Networks

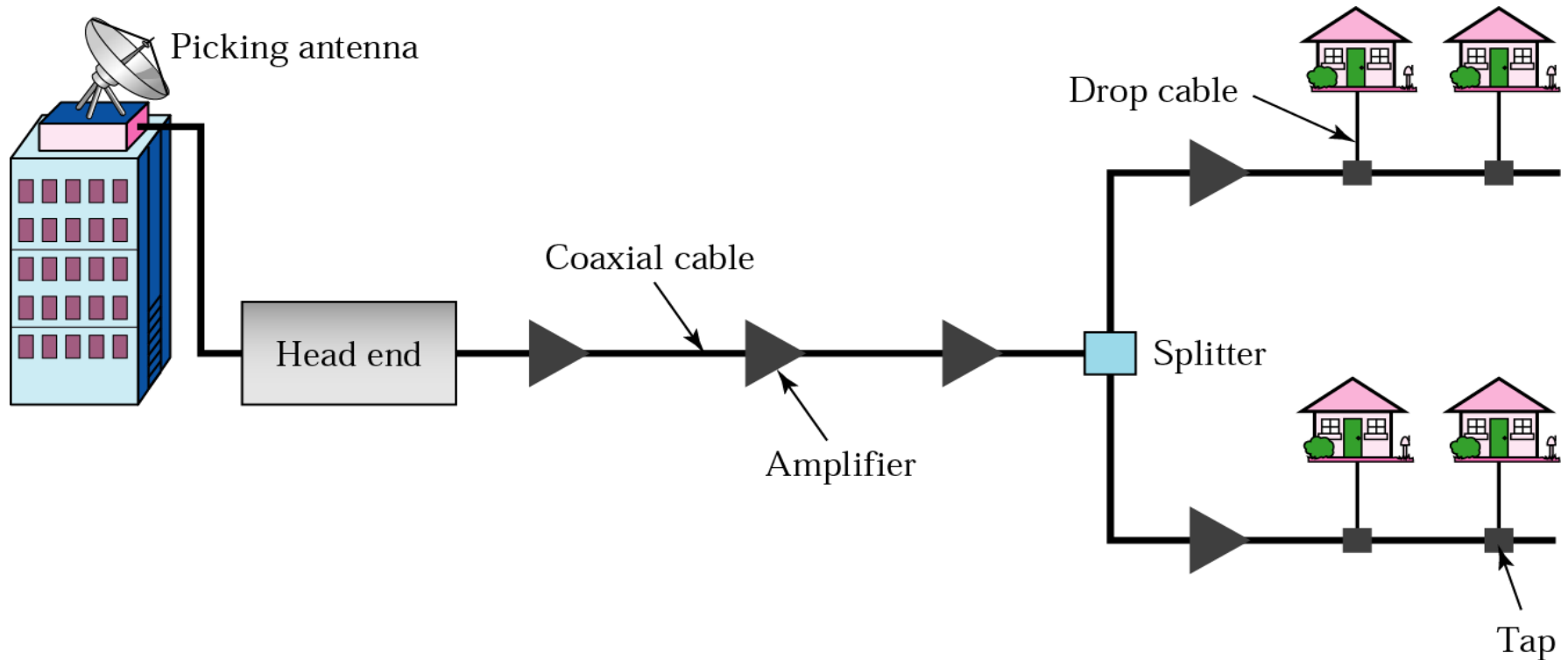
Hybrid Fiber-Coaxial (HFC) Network



Cable Modem

❑ Traditional cable Networks

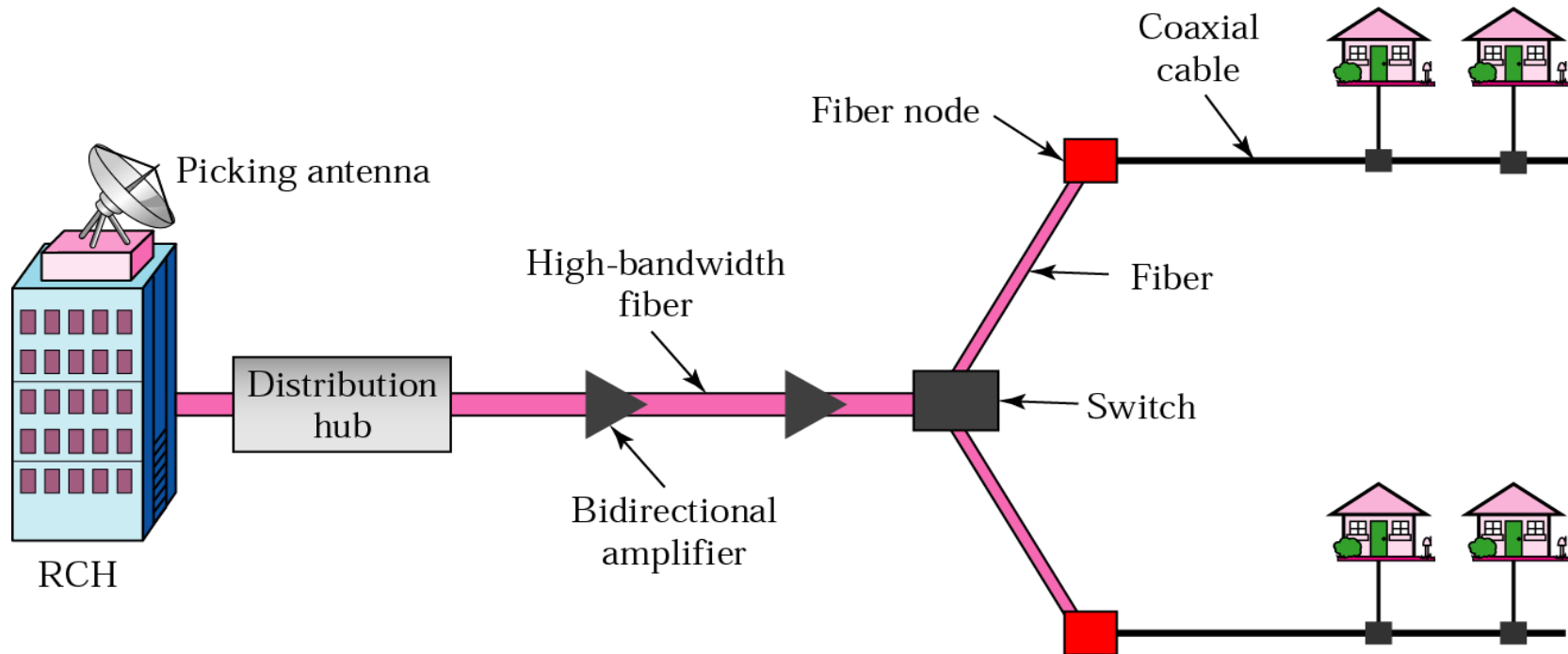
❖ community antenna TV (CATV)



❑ Communication in the traditional cable TV network is unidirectional.

Cable Modem

❑ HFC Network



- ❖ **RCH** : Regional cable head; serving 400,000 subscribers;
- ❖ **Distribution hub**: serving 40,000 subscribers
- ❖ **Coaxial cable** : serving 1,000 subscribers
- ❖ **Communication in HFC cable TV network can be bidirectional.**

Cable Modem

Note

**Communication in an HFC cable TV network
can be bidirectional.**



9.5 CABLE TV FOR DATA TRANSFER

Cable companies are now competing with telephone companies for the residential customer who wants high-speed data transfer. In this section, we briefly discuss this technology.

Topics discussed in this section:

Bandwidth

Sharing

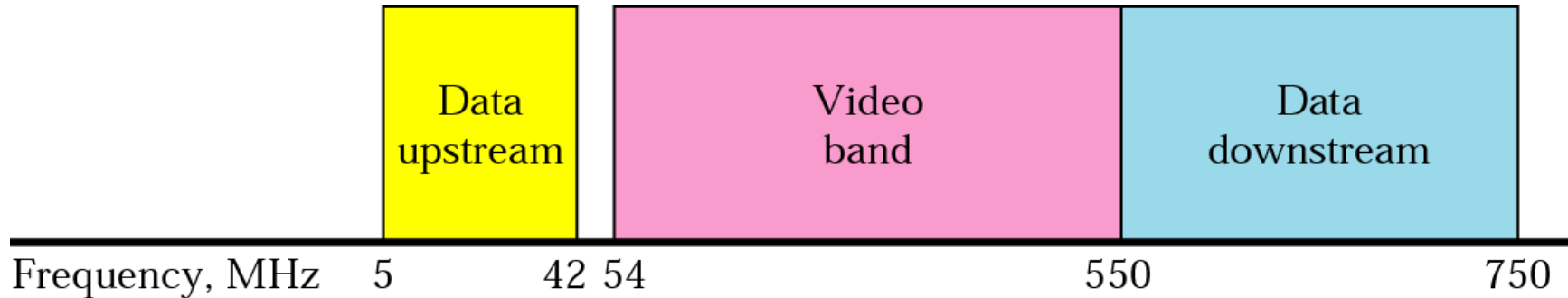
CM and CMTS

Data Transmission Schemes: DOCSIS



Cable Modem

Bandwidth



Video band

❖ 54 to 550 MHz

❖ TV channels : 6 Mhz x 80 channels



Cable Modem

❏ Data downstream band : dividing into 6Mhz channels

❖ Modulation

- Downstream data are modulated using 64-QAM

❖ Data rate

- 6 bits for each baud in 64-QAM (1bit : control bit)
- Theoretically, $5\text{bits/Hz} \times 6\text{ Mhz} = 30\text{ Mbps}$

❏ Upstream data band

❖ Modulation

- upstream data band uses lower frequencies that are more susceptible to noise and interference
- for this reason, using QPSK instead of QAM

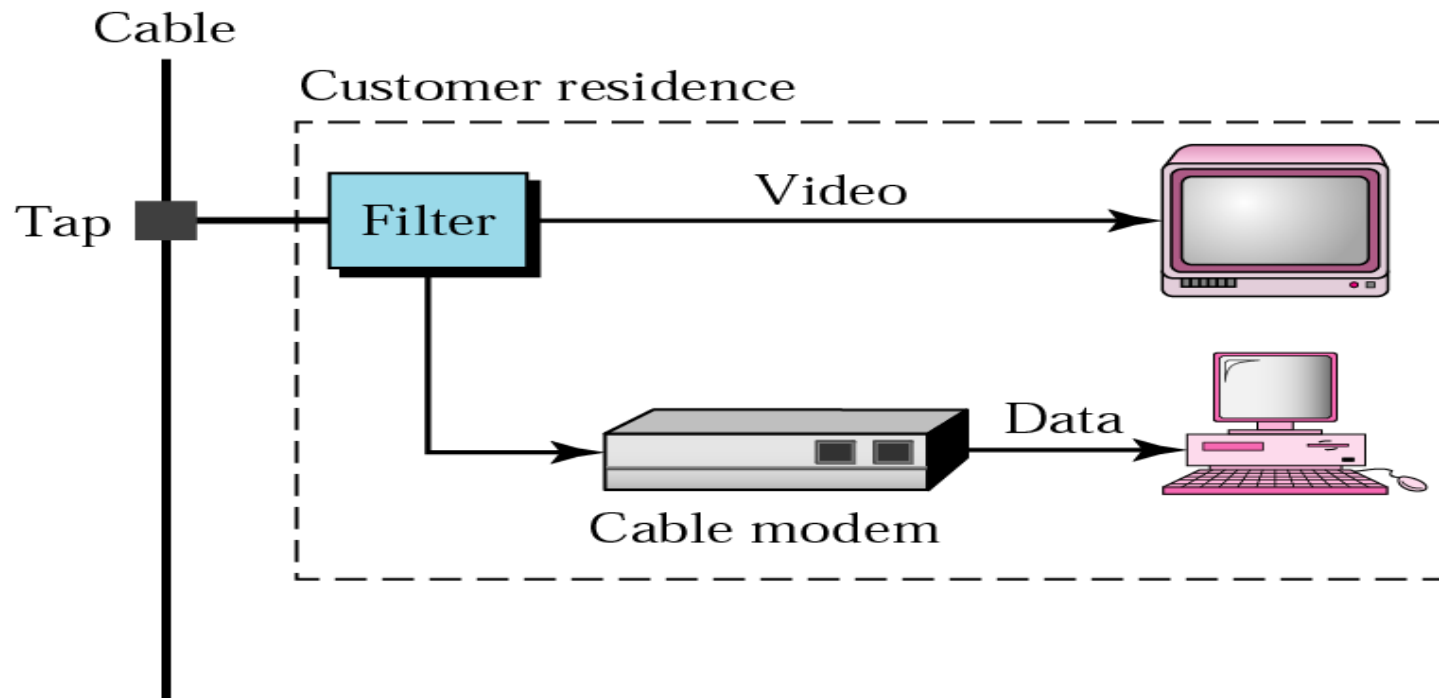
❖ Theoretical data rate : $2\text{ bits/hz} \times 6\text{ Mhz} = 12\text{ Mbps}$



Cable Modem

- ❑ The Cable Modem (CM) is installed on the subscriber premises.

Figure 9.17 *Cable modem (CM)*

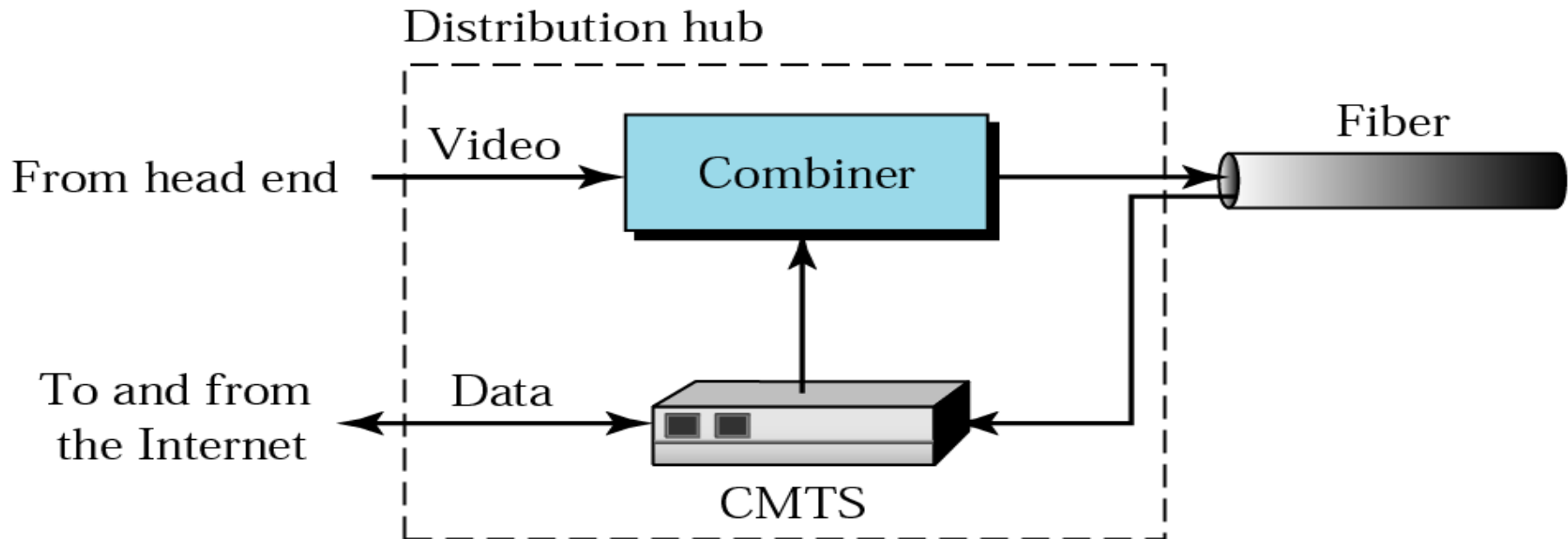


Cable Modem

❑ Cable modem transmission system (CMTS)

- ❖ The CMTS is installed inside the distribution hub by the cable company.

Figure 9.18 *Cable modem transmission system (CMTS)*



Data transmission Schemes (DOCIS)

❑ **DOCIS (Data Over Cable System Interface Specification)** defines all the protocols necessary to transport data from a CMTS to a CM.

❖ **Upstream Communication**

- It describes the steps that must be followed by a CM.

❖ **Downstream Communication**

- There is no contention because there is only one sender.
- The CMTS sends the packet with the address of the receiving CM, using the allocated downstream channel.



Summary (1)

- ❑ The telephone, which is referred to as the plain old telephone system (POTS), was originally an analog system. During the last decade, the telephone network has undergone many technical changes. The network is now digital as well as analog.
- ❑ The telephone network is made of three major components: **local loops, trunks, and switching offices**. It has several levels of switching offers such as end offices, tandem offices, and regional offices.

Summary (2)

- ❑ Telephone companies developed another technology, digital subscriber line (DSL), to provide higher-speed access to the Internet
- ❑ Community antenna TV (CATV) was originally designed to provide video services for the community. The traditional cable TV system used coaxial cable end to end. The second generation of cable networks is called a hybrid fiber-coaxial (HFC) network. The network uses a combination of fiber-optic and coaxial cable.



Q & A

